Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	11 October 2025
Team ID	SWUID20250177148
Project Name	SB Works
Maximum Marks	4 Marks

Technical Architecture

The technical architecture of SB Works follows a scalable 3-tier client-server model. This structure is composed of a presentation layer (frontend), a business logic layer (backend), and a data storage layer (database). The solution is built on the MERN stack (MongoDB, Express.js, React, Node.js) to facilitate real-time communication, efficient data exchange, and a seamless, immersive experience for users.

Table-1 : Components & Technologies:

S.No.	Component	Description	Technology
1	User Interface (Frontend)	A web-based, real-time, and visually appealing interface for clients, freelancers, and admins. It serves as the presentation layer and communicates with the backend via RESTful APIs.	React.js, Socket.IO, Bootstrap, Material UI
2	Application Logic (Backend)	The server-side logic that manages requests and responses efficiently. It handles user management, project actions, and communication, providing a robust foundation for the application.	Node.js, Express.js
3	Database	A scalable and efficient solution for storing and retrieving all application data, including user profiles, projects, applications, and chat messages.	MongoDB, Mongoose

Table-2: Application Characteristics:

S.No.	Characteristic	Description	Technology Stack
1	Open-Source Frameworks	The platform is built using a modern, open-source stack for both the client and server sides.	React.js, Node.js, Express.js, Bootstrap, Material UI
2	Scalable Architecture	The application uses a 3-tier client-server architecture with RESTful APIs, ensuring efficient data handling and the ability to scale as user demand grows.	MERN Stack (MongoDB, Express.js, React, Node.js)